



CYS 402 – Mathematical Foundations of Cyber Security

Mathematical models to detect robot users for website services

Student name
Bayan Abu Ribiyah
Dana Alahmari
Jawaher Alamer
Norah Alsubaie

Supervised by

Dr. Naya Nagy



Abstract:

This project focuses on identifying bots within website services by analyzing various mathematical models. It examines user behavior patterns such as mouse movements, session durations, and click rates. The techniques evaluated include Bayesian networks, CAPTCHA, and device fingerprinting, which are employed to distinguish between legitimate users and bots. The study provides insights into optimizing these methods to improve bot detection accuracy, enhance user experience, and ensure better security, by combining probabilistic models and behavioral analysis.

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1.1 Introduction:

Due to an increase in the usage of automatic bots in web services, it is very important for service providers to know which of the users accessing their website service is robot. The work focuses on analyzing mathematical models that provide a structured methods to distinguish between legitimate or real users and automated programs or robots by analyzing behavioral, structural, and temporal data. In contrast to humans, robots often display unnatural characteristics such as extremely fast response times, repetitive click sequences, or predictable browsing patterns. Techniques such as probabilistic reasoning with Bayesian networks, convolutional neural networks (CNNs), and CAPTCHA systems, device fingerprinting, and query rate analysis serve as important complementary methods. These models strengthen website security and help maintain user trust, protect sensitive information, and reduce risks such as fraud, data scraping, and system overload like DoS attack.

Recent research shows how bots accessing website problem across different domines. according to **Imperva (2024)**, automated bots have seriously disrupted e-commerce operations by performing tasks like inventory denial, account takeover, and data scraping, leading to big financial losses and reduced trust [1]. The **2023 Imperva Bad Bot Report** found that nearly half of global web traffic is generated by bots, the majority of which are malicious [2]. Beyond e-commerce, **Lobato and Oliveira (2020)** examined how bots penetrate social and interaction networks to manipulate engagement and public opinion [3]. While **Tünnermann, Schwemer, and Teichmann** showed that even a tiny proportion of social bots can affect overall public sentiment [4]. Together, these findings support the significance of creating intelligent detection models such as those covered in this work to ensure safer, more reliable online systems.

1.2 Motivation:

Why We Chose Bot Detection?

Every time we check a box that says "I'm not a robot," we wonder: how does the internet really know?

That simple question sparked our curiosity and led us here. We chose this project because we wanted to learn what happens behind that checkbox—to understand the quiet math that separates humans from machines. It wasn't just about security or algorithms; it was about answering a small, everyday wonder with something real. And that curiosity became our guide.



1.3 Problem Statement:

Modern websites are increasingly targeted by sophisticated bots that closely imitate real user behavior, leading to data theft, system overload, and financial losses. While many bot detection tools are currently in use, their effectiveness against advanced bots remains limited. The main problem lies in the lack of a clear analytical evaluation of existing detection tools to assess their strengths, weaknesses, and readiness for future threats. Therefore, this project focuses on analyzing current bot detection methods to support future development and improvement.

2.Literature review:

[1] How Bots Affect E-commerce

Research on automated activity across the web shows that bots have become a serious issue for different types of online platforms, but the problem appears most clearly in e-commerce environments. One study reports that about 17.7% of traffic on online stores comes from malicious bots, and in some extreme cases the percentage is far higher—sometimes over 90% of the website’s activity isn’t human at all.

[2] Bad bot report

These bots don’t all behave the same. Some monitor prices on competing sites, others race to buy limited-stock products before real customers can even load the page, and many target user accounts for fraud. What makes this worse is that such traffic increases server costs and damages analytics, so companies end up making decisions based on distorted data.

A more recent source adds to this picture. According to the 2023 Imperva Bad Bot Report, nearly one-third of all web traffic is generated by harmful automation, and these bots are now smarter, harder to detect, and able to blend in with regular users much more easily.



[3] Bots in Social and Interaction Networks: Detection and Impact Estimation

Many of them use residential proxies or spoofed devices, making it difficult for even advanced systems to block them. E-commerce platforms remain top targets and constantly face scraping, credential stuffing, and automated checkout bots. The report also highlights that account takeovers are increasing because attackers rely on huge collections of leaked credentials. Traditional defenses like simple IP blocking just don't work anymore because bot operators rotate identities so quickly.

A similar pattern appears on social media, where bots have evolved from simple spam accounts into sophisticated automated profiles. Early research used traps like honeypots or basic behavioral cues to spot bots, but these methods became less reliable as bots improved. Later studies shifted toward analyzing how accounts connect with one another. Graph-based techniques revealed that bots often form clusters with synchronized behavior that looks harmless individually but becomes suspicious when viewed at the network level.

[4] ARE SOCIAL BOTS A REAL THREAT?

Research also shows that these bots play a noticeable role in spreading misinformation, especially during elections, crises, and public health emergencies. Even though some bots are harmless—like those summarizing news or assisting with basic tasks—the general agreement in the literature is that the harmful ones dominate the landscape and pose a meaningful risk to public opinion

[5] CAPTCHA Types and Breaking Techniques: Design Issues, Challenges, and Future Research Directions

One of the most common tools used to separate humans from bots is CAPTCHA, introduced in 2000. Over time, however, advances in machine learning made older CAPTCHA methods less effective. Studies categorize them broadly into OCR-based and non-OCR-based designs. OCR-based CAPTCHAs are faster and easier for users, but they are also the easiest for bots to break. Non-OCR types tend to be more secure but often frustrate users and are less accessible, especially on smaller devices or for individuals with disabilities



[6] Web robot detection: A probabilistic reasoning approach

Attack techniques have improved too, moving from simple image splitting to deep-learning models that can solve many CAPTCHA variations with surprising accuracy. This pushes researchers to find better ways to balance security with usability.

Past robotic activity detection relied on blacklisting IP ranges or identifying specific user agents, but these strategies quickly became outdated as crawlers learned to rotate their identities. Newer approaches focus on how users navigate a site: their click rates, session lengths, and general browsing speed. Probabilistic models, including those built on Bayesian networks, have shown strong potential because they can handle uncertainty and combine multiple behavioral signals at once

[7] Web robot detection techniques: Overview and limitations - ResearchGate

Previously, detection methods for web robots primarily depended on static rules like the use of black IP addresses or predetermined server log patterns. Although the methods were easy to apply, they lost effectiveness quickly due to the rapid development of the automated agents. Current bots are capable of doing IP address rotation, imitating real users' browsing actions, and even changing their behavior in response to detection attempts, thus making the static defenses ineffective. Literature reviews in this field now talk about the need for a new generation of bot-detection systems that are adaptive and data-driven instead of those that were fixed rules based. This kind of analysis entails real-time checking of a broad range of behavioral indicators like the users' navigation routes, the length of their sessions, the rates of their requests and their click patterns. Through the integration of these signals, automated systems can spot abnormal or dubious actions even in cases where the bots are trying their hardest to look human. The move to adaptive models illustrates the evolving character of automated threats: the more skilled the bots become, the more flexible the detection methods must be as well, they must be able to learn new traffic patterns, and continuously update their decision-making criteria for accuracy and reliability.



[8] CAPTCHA Types and Breaking Techniques: Design Issues, Challenges, and Future Research Directions

For quite a long time, CAPTCHA has been the primary method that was relied upon to tell humans apart from bots. Notwithstanding, recent studies reveal that the said protection has lost much of its strength over time. Through this research, the authors provide a detailed account of how the early text-based and image-based CAPTCHAs have gradually become ineffective due to the large strides made in machine learning, optical character recognition, and deep learning attacks. What has been an operation only for human intelligence is now being handled with great accuracy by automated systems. Besides the security vulnerabilities, the research outlines the difficulties that CAPTCHAs may present to the users, especially the disabled people. Thus, the authors propose that verification methods in the future should lean more towards adaptive and behavior-based systems that are secure as well as user-friendly. This indicates that the role of CAPTCHA is evolving from an exclusive solution to just one element of a larger detection strategy.

[9] Web Robot Detection Techniques: Overview and Limitations

As CAPTCHA becomes more and more difficult to challenge due to its inherent ease of solution, the focus has now turned to wider bot detection techniques. This research paper provides a comprehensive account of the main different detection methodologies that have been in use over the years, which range from rule-based systems, traffic pattern analysis, to behavior-based approaches. Although these techniques were practically effective against the simple bots, the authors of the paper fluently reason that they have lost their reliability when it comes to the modern types of bots that exhibit very similar characteristics to humans. One of the most significant issues raised is the large volume of false positives, which leads to the problem of legitimate users being incorrectly blocked. The paper also states that the conventional detection systems cannot be up to pace with the rapidly changing strategies of the bots.



[10] Bot Recognition in a Web Store Using Unsupervised Learning

Researchers have started to search for solutions based on machine learning to tackle the problems of static detection. This paper reveals a way of unclassified learning that tells apart bots in e-commerce sites with no need for labeled data. Treating user behavior patterns and clustering techniques to detect anomalies, the system, in contrast to the traditional rule-based systems, achieves a significantly higher accuracy rate in identifying humans and bots. The results demonstrate convincingly the advantages of the new method in terms of the detection performance, particularly in the case of real-world e-commerce environments. Nevertheless, the authors are also concerned with the issues of feature selection and the model's capacity to generalize across different datasets. Regardless of these limitations, the research emphasizes the huge potential of unsupervised learning as a practical solution when labeled bot data is hard to come by.

[11] Bot Detection in Social Networks Based on Multilayered Deep Learning

Unsupervised learning is good in the case of online stores; meanwhile, social media environments require even more sophisticated detection methods. The paper presents a multilayered deep learning model that is capable of detecting social bots by analyzing posting behavior, interaction patterns, and content features. The results reveal very high detection accuracy even when bots resort to using sophisticated evasion techniques to remain undetected. This indicates that deep learning technology is among the most powerful tools in large-scale bot detection. Yet, the study also brings to light some significant obstacles, including the high computational cost as well as the opaqueness of the decision-making process in these models. Regardless of these issues, the research overwhelmingly backs the role of deep learning in managing complex and extensive bot activities on social media.



[12] Demystifying Website Traffic Bots: Understanding Their Role and Impact on Analytics

Apart from the detection methods, the scholars have also tried to find out the exact impact of the bots on the websites. The present article categorizes the website bots in the way that good bots like search engine crawlers are separated from bad bots engaging in web scraping, fraud, and credential stuffing. It describes the way the malicious bots utilize the server resources, degrade the performance of the websites, and interfere with the generation of user behavior data. Consequently, the companies do not realize but the data generated is often out of sync with the actual and they make wrong decisions regarding their strategy. This piece of work makes it clear that bots are not only a cybersecurity issue but also an indirect trial of business performance and the trustworthiness of data.

[13] Bot Detection | How to Detect Bots (Radware)

The document describes the real-world bot detection from the industry cybersecurity angle. Modern bots have been explained as being responsible for the whole range of malicious activities from account takeover through credential stuffing, scalping, and even automated fraud. The article notes the incapacity of the traditional rule-based security systems to deal with these types of modern threats. It rather, highlights behavioral analysis, AI, and challenge-response mechanisms as the major detection techniques. This research assures that the modern bot attacks are extremely dynamic and need an intelligent, adaptive, and multi-layered defense systems instead of static security rules.

[14] What Is a Device Fingerprint and How Is It Used? (Incognia)

The piece defines device fingerprinting as an innovative method of identification that relies on the assembly of browser configurations, device features, network usage patterns, and operating system attributes to create a user profile. In contrast to cookies, device fingerprinting continues to function even when users are trying to mask their identity, thus making it a very effective tool



for bot detection and fraud prevention. The research has also looked into some of the associated privacy and legal issues especially in systems that rely on gathering detailed behavioral data. This source appreciates the deployment of device fingerprinting as a strong and auxiliary security layer that fattens bot detection accuracy without necessitating direct user interaction.



3.Preliminary experiments:

Experiment 1: Probabilistic Reasoning Using Bayesian Network

Using Bayesian Network to predicate if the user bot or human based on mathematical probability derived from session features:

The number of clicks per minute

Session duration by seconds

Percentage of image requests

Accessing (Robots.txt file) 1=True, 0=False

Calculating the probability of session being robotic by Bayes:

$$P(\text{Bot} \mid F1, F2, F3, F4)$$

The Naïve Bayes classifier was implemented in Python to test the database.

If the sessions have high click rates, session duration is low, and repeated requests, this could indicate that the user is a bot. For example, if the user clicks very quickly or visits the robots.txt file repeatedly, this is a strong indication of bot behavior. By satisfying these features, the probability that the user is a bot exceeds 80%. [6].



```
1 from sklearn.naive_bayes import GaussianNB
2 import numpy as np
3
4 X = np.array([
5     [20, 50, 0.6, 1, 0.1],
6     [3, 600, 0.1, 0, 2.5],
7     [15, 80, 0.4, 1, 0.2],
8     [5, 720, 0.2, 0, 1.8]
9 ])
10
11 y = np.array([1, 0, 1, 0])
12
13 model = GaussianNB()
14 model.fit(X, y)
15
16 new_session = np.array([[18, 60, 0.5, 1, 0.15]])
17 proba = model.predict_proba(new_session)[0][1]
18
19 print("Bot probability:", round(proba, 3))
20 print("Result:", "Bot" if proba >= 0.8 else "Human")
21
```

A basic example of probabilistic reasoning in this code involves the application of a **Gaussian Naive Bayes classifier** to forecast a bot or human session based on the session. The model takes advantage of a number of behavioural characteristics such as **Average Time Between Clicks** as bots tend to click with a lot more speed and frequency than human users.

Libraries

- **NumPy** is employed to make numerical arrays of the data.
- **GaussianNB** is scikit-learn machine-learning classifier: it computes the probability based on a normal distribution of the data.

Dataset

- Some of the session features of **X** include click rate, session duration, image request percentage, robots.txt access and average time spent on clicks.
- **y** represents the labels (1 = bot, 0 = human).

```
X = [...]  
y = [...]
```

Training the Model

```
model = GaussianNB()  
model.fit(X, y)
```

The pattern defining the difference between bots and humans is learned by the classifier.

Prediction

```
new_session = [...]  
proba = model.predict_proba(new_session)[0][1]
```



The model runs the probability of the new session to be a bot.

Experiment 2: Query-Rate Pattern Detection

Detect the automated behavior by observing query timing pattern. Ten virtual sessions were generated and focused on the timestamps of every request.

The sessions that had time gap of less than one second were classified as robotic sessions. Because for human user's sessions there was time gap between two and 15 seconds.

This method is distinguished by lightweight, based on databases, and simple to conduct, hence appropriate for real time monitoring applications as well [7].

Although the query – rate detection methods is efficient and simple, it suffers from limited accuracy because it relies solely on request timing . it cannot handle sophisticated bots that mimic human behavior or adapt to varying session patterns.

Therefore, this approach is suitable only for simple, real-time detection scenarios and not for advanced or large-scale systems.

Experiment 3: Combining CAPTCHA and Behavioural Model

The third experiment is an attempt to combine a CAPTCHA and a behavioural model to enhance the overall accuracy of bot detection by comparing the accuracy results obtained with and without presence of the CAPTCHA element in the combined response time measure. The methodology consists of an artificial dataset of 10 participants, 5 human subjects and 5 automated agents, with the performance of each participant being the CAPTCHA performance (0 if failed, 1 if successful) and the time in seconds. The probability that a participant will be defined as a bot has been estimated from the formula: $\text{BotProbability} = 0.7 * (1 - \text{CAPTCHA score}) + 0.3 * (1 / \text{response time})$, which combines both verification and behavioural variables in order to simulate real user profiles. This combination methodology is expected to have less false positive than single models, using either CAPTCHA or



behavioural analysis only. This is based on the premise that genuine users will have longer and more variable response time and a higher rate of CAPTCHA success whereas bots will have shorter completion time and a lower CAPTCHA score[5].

CAPTCHA systems have both advantages and disadvantages. They help block automated access and are easy to add to websites however, their protection is getting weaker against AI attacks , they can annoy real users when too hard ,and most design are static and not adaptive to new threats.

3 Discussion and Analysis

3.1 Discussion and Analysis

The mathematical models used to identify the robot users in the website services have been fully discussed in this section. The experiments carried out: probabilistic reasoning based on Bayesian networks, query rate pattern detection and a hybrid CAPTCHA-behavioral model are all directed to the goal of distinguishing between bots and human users on the basis of structural and temporal data.

3.1.1 Probabilistic Reasoning based on Bayesian Network.

Bayesian Network model incorporates its major features of the session like maximum number of rate of clicking, duration of the session, percentage of the image requesting, and access to robots.txt. Sessions that had a high rate of clicks, less time consuming, and a repeat of an attempt to access robots.txt had a bot probability higher than 80. This proves the ability of the model to manage uncertainty and integrate a number of indicators. Nevertheless, intelligent bots with the ability to mimic human-like design present a drawback to this technique.[9]

3.1.2 Query rate pattern Detection.



In this experiment, the shortest delay between successive user requests was studied. The range of natural delays is usually 2-15 seconds with human users and the bots take under one second. This renders query rate analysis to be lightweight and effective. They can not be detected, though, since artificial delays can be introduced by bots, making the technique less reliable when applied on its own.

3.1.3 Hybrid CAPTCHA and Behavioural Model.

The success of CAPTCHA and response time were weighted in the hybrid model and a weighted equation was used. CAPTCHA performance was considered 70 percent weighted and response speed 30 percent. This has decreased the number of false positives and improved accuracy because human users have been shown to have higher CAPTCHA success rates and longer and more varying response times. Bots, on the contrary, either manage to pass CAPTCHA or do it extremely fast.

3.1.4 Evaluation of Methods

- Bayesian model demonstrated a good predictive performance though it is vulnerable to the advanced level of bot behavior.
- Query pattern technique was effective but was too basic to identify sophisticated bots.

The hybrid CAPTCHA-behavioral model also offered the best accuracy with the use of two independent indicators: human verification and behavioral timing. components allows detecting the bot in a more resilient way than an individual approach.

3.2 OCR-Based CAPTCHA Schemes



CAPTCHA schemes with OCR give the user distorted alpha numerical characters, which they are required to key in correctly in order to continue. Such distortions like irregular shapes, noise, and lines are meant to make sure that the automated systems do not identify the characters.[8]

Strengths:

- Easy and quick to use by human beings.
- Simple to develop and put in place.
- Uses insignificant computing resources.[8]

Weaknesses:

- OCR, deep learning, and segmentation algorithms are capable of recognizing distorted text very accurately today.
- May be tough or unreachable to those with visual impairment.
- CAPTCHA that is overly distorted makes usability less due to the frustration experienced by legitimate users.[8]

Conclusion:

CAPTCHA based on OCR is not adequate in the context of the modern environment to ensure the prevention of bots. Its efficiency is enhanced greatly when combined with behavioral models or probabilistic detection methods, including those that were used in the experiments presented above.[8]



3.3 Non-OCR-Based CAPTCHA Schemes

CAPTCHA schemes that are not based on OCR do not entail reading deformed text in the form of a challenge. Rather they assess the user based on visual recognition, object selection, puzzle solving or interaction patterns that humans can perform very easy, but can hardly be mimicked by automated bots. These schemes are intended to be used to get around the weaknesses of OCR-based CAPTCHA by moving towards character recognition to cognitive and behavioral challenges that are unassailable by modern deep learning and segmentation attacks.[10]

Strengths:

- Make it more resistant to automated attacks as bots are weak in the field of cognitive and perceptual tasks.
- Easier to evade by deep learning models than distorted-text CAPTCHA.
- Can have interactive interfaces (e.g. dragging, image selectiCanannot be easily imitated by bots.[8]

Weaknesses:

- Images or interactivity use up more storage, processing or bandwidth by May.
- Are able to compromise usability to certain users, those with disabilities or slow computers.
- Even sophisticated AI systems are capable of being trained to solve some of them, in particular, image-based CAPTCHA.[8]

Conclusion:

CAPTCHA schemes that are non-OCR-based are more secure than the traditional OCR-based schemes in that the scheme addresses perceptual and behavioral issues instead of considering text recognition. Nevertheless, they are effective only when



there is a balance in terms of robustness and usability. They can be used together with behavioral models, e.g. query-rate analysis, and adaptive detection, which was applied in the experiments, to offer more robust and effective defense against contemporary automated bots.[10]

3.4 Detection Through Query Rate Patterns

The query rate pattern is used to detect human or bot actions, the time interval between consecutive user action during a session is analyzed to identify whether the user is behaving like an automated bot or a human. The approach is based on the premise that human users will intuitively spend more and more erratic amounts of time between actions whereas bots delay their requests at regular and more rapid intervals.

Under this method, the system will log the times of every request under a specific session and the point where the time difference between actions is smallest and the average time between actions are computed. The request sessions which show very minimal time delays (less than a second between successive queries) reflect automated behavior, as humans need time to read, contemplate and type or navigate to post another request. Human sessions on the other hand tend to exhibit bigger and more disorganized intervals with a span of several seconds to over a decade.

Through these timing features, the model is able to tell between those sessions that act unnaturally fast and those that follow the patterns of human interaction. It is lightweight, no extra effort by the user is needed and can be run passively in the background using existing timestamp logs. Thus it can be appropriately used in real time monitoring, and it can effectively detect bot activity without involving complex calculation and further verification.

All in all, query-rate patterns analysis offers a very simple, yet useful, behavioral feature to use in detecting bots, particularly when there are cases where the bots are



trying to behave like normal human navigation and yet still not able to generate natural variation of human timing.

3.5 Concluding remark:

In this work, the main goal was to tell apart the human and robot users on the web by means of mathematics and engagement models. In the beginning, the methods of detection concentrated on rather primitive probabilistic techniques, mainly Bayesian inference on the basis of clicks and logging patterns. Eventually, these techniques turned out to be pretty powerful and hence, they were combined with behavior-based and temporal and structural features as well as adaptive strategies, resulting in a detection system that is more flexible and accurate at the same time.

In a similar way, CAPTCHA has gradually improved along with these detection systems. The text-based challenges of the past are now replaced by CAPTCHAs that recognize images, analyze users' behavioral interactions, and regulate difficulty levels in correspondence with the degree of detected suspiciousness. This progress keeps pace with the evolution of bot detection starting with basic models moving up to multi-layered, behavior-aware, and machine-learning-driven approaches, thus reflecting the increasing sophistication of automated attacks.

When it comes to future research, a few directions can be pointed out. Feature extraction, real-time adaptive detection, hybrid modeling with machine learning, and ensemble techniques will all work towards the improvement of bot detection systems. Not only that but also spreading these methods among mobile applications, cloud services, and IoT environments will undoubtedly have a bigger impact. The use of dynamic and layered frameworks that merge probabilistic, behavioral, and CAPTCHA-based analysis will at the same time fortify security and enhance the user experience.

Eventually, the combination of intelligent CAPTCHA systems with the latest bot-detection models represents a mixed security approach for the internet. By utilizing probabilistic analysis, real-time behavior monitoring, and a responsive challenge-response system, the new generation of systems will be able to resist the emerging bot tactics and be still friendlier to the user at the same time.



3.6 Suggestions:

To enhance the detection methods for bots in the future, several areas are suggested to be further developed. The first one is to improve the quality of the extraction of behavioral features, which is a key task in our project to a large extent, through the analysis of user interactions, such as click frequency, session duration, and response speed. Future research should allow for a wider range of behavioral indicators to be gathered, and among those, mouse-trajectory patterns, scroll behavior, idle time distribution, and inconsistency metrics can be presented as sources that provide a more credible separation between human actions and robotic scripts.

The other direction worth mentioning is combining advanced machine-learning techniques with the Bayesian and query-rate ones established in this work. Algorithms like decision trees, support vector machines, or even small-scale neural networks may be instrumental in identifying sophisticated bots that are trying to imitate human behavior. The application of hybrid learning models could result in the detection being more sensitive without the trade-off of system performance.

Another point to be considered is real-time monitoring that should take precedence. The majority of the bots do so according to the environment and therefore, one way to increase accuracy is to implement dynamic thresholds that get updated depending on the live traffic, e.g., click-rate limits varying based on peak or off-peak hours could help in reducing false positives and at the same time the system would be automatically adapting to new wild bot strategies.

Furthermore, the CAPTCHA + behavioral analysis model could undergo a transformation through the implementation of adaptive challenge levels. Rather than exposing every user to the same CAPTCHA, the system might alter the difficulty based on the perceived risk. Users who are determined to be at high risk will be given very challenging tasks, while low-risk users will be only slightly disturbed. The proposed method would hence be kind to the users while at the same time secure.

It is furthermore recommended that subsequent research examine multi-layered detection frameworks, wherein Bayesian probability, request-timing analysis, and CAPTCHA performance are all integrated into one scoring engine. This would result in a stronger decision mechanism that would be able to unearth both the basic and the advanced bots.

Moreover, it would be a great step if the techniques developed in this work were applied to different settings such as mobile apps, cloud services, and IoT platforms, thus making them more



widely usable. A joint effort by researchers and industry partners can be an additional factor in providing support for the development of universal and scalable bot-detection solutions.

This way, the coming studies can aid in the formation of bot-detection systems that are more precise, flexible, and user-friendly, thus being able to counteract the increasingly sophisticated automated threats.

4. Conclusion:

4.1 introduction:

We showed in this project that it is difficult to identify and classify the robots in website services, especially with bots using machine learning and AI tools. So, it is very important to create mechanisms and systems to detect bots within online services. CAPTCHA is a very critical system to detect robots, which, in the past, was only easy text-based puzzles to solve, but now, according to the improvements of bots, it is becoming necessary to combine the enhanced mathematical algorithms into CAPTCHA systems to determine whether the user is authorized or a bot.

4.2 Findings and Contributions:

In order to improve detection accuracy, we can combine machine learning techniques, statistics, and mathematical models to improve the CAPTCHA system to restrict bots from cracking, while users can access the online service easily. By analyzing these three methods, CAPTCHA can be optimized to detect robots that rely on AI and machine learning mechanisms.

1. Recognizing Natural Human Behavior Online

When it comes to distinguishing between real people and automated programs online, much of it comes down to observing how someone actually uses a website. For example, when you browse, your mouse likely doesn't move in straight lines; you might scroll quickly through some parts and slowly through others, and you might make the occasional small mistake while typing. These small imperfections and inconsistencies are completely normal. They are signs of a real person thinking, deciding, and perhaps getting slightly distracted [6][7].

On the other hand, automated bots are built for a specific task. They follow their instructions with a level of speed and consistency that just doesn't appear human. Every action is precise,



every movement follows a pattern, and there are no hesitant pauses or corrections that people naturally make. Due to this tell-tale pattern, many websites now use specialized background tools that focus on these subtle behaviors. They don't monitor what you do so much as **how** you do it, looking for that unnatural robotic rhythm in clicks, scrolls, and keystrokes [13].

2. The Concept of a Device's Digital Fingerprint

Another method that has become essential involves creating a kind of **digital profile** for the device visiting a site. This is not just one piece of information, but a combination of many technical details that together form a **unique signature** [14]. It includes things like the exact browser model, screen size, list of installed software add-ons, and even the device's time zone settings.

The value of this fingerprint lies in its **stability and uniqueness**. For a real person using their own computer or phone, this profile remains relatively constant. However, if a website notices that the same digital fingerprint suddenly starts behaving in an impossible way like loading hundreds of pages per second or if the fingerprint itself keeps changing erratically, it raises a significant red flag. This technique is incredibly useful for spotting **advanced bots** that try to hide by constantly changing their network address because the core identity of the device itself is much harder to disguise effectively [14].

3. Employing Smart, Learning Detection Systems

As bots have become more sophisticated, the systems designed to catch them have had to evolve as well. This is where **intelligent, learning-based technology** comes into play. These systems are trained by analyzing large amounts of data from confirmed human users and known bot attacks. Over time, they learn to detect the incredibly subtle clues and complex patterns that might indicate an automated program even one designed to mimic human behavior [10] [11].

Some of these models work with clear examples, improving their ability to recognize known threats. Others are designed to hunt for anything unusual or "out of place" in the data, helping to uncover entirely new types of bots. The most advanced versions can analyze the **entire sequence of a website visit**, including the order of pages viewed, the time spent on each, and the flow of mouse movements, to catch imposters that might bypass simpler checks. The key is that these systems **never stop learning**. As they encounter new traffic, they adapt, which is crucial for staying ahead of the constantly evolving bot tactics[13].

4. Why a Single Solution Is No Longer Enough

In today's landscape, relying on just one method to stop bots is like locking only the front door of a house. Determined attackers will find another way in. Bots are constantly being rewritten



and refined to bypass specific defenses. Therefore, the most effective approach is to build a **multi-layered security system**[13].

This means not only using one technique, but thoughtfully combining several. A website might first check the device's fingerprint[14], then analyze the user's behavior as they browse, and finally use an intelligent learning model to score the risk of the entire session[10][11]. Each layer acts as a **separate checkpoint**, making it exponentially harder for bots to slip through undetected. A bot clever enough to bypass the first check may still be caught by the second or third. This "defense-in-depth" strategy makes it far more difficult for automated threats to succeed, as they must now perfectly mimic human behavior across multiple independent tests simultaneously[13].

5. Modern Verification: Smarter Challenges for Smoother Visits

Most internet users are familiar with **CAPTCHAs** those small tests asking you to identify traffic lights or type distorted text. Their core purpose has always been to present a simple puzzle for humans that is difficult for machines to solve. However, their design has evolved significantly. Modern versions are far more **intuitive** and **user-friendly**. Many now use **image-based puzzles** that are faster for people to solve. More importantly, the latest systems are **risk-based**, meaning they often run invisibly in the background, presenting a visible challenge only if the user's activity seems suspicious based on other detection layers. Some systems even analyze how you interact with the puzzle itself such as the path of your mouse as you click to verify humanity before you even finish the task.

When integrated into a broader, multi-layered defense strategy, these challenge systems become a **precise tool**. They add a final, **flexible layer of security** that activates only when needed, effectively blocking bots while minimizing disruption for legitimate visitors, leading to a **safer and smoother experience** for everyone.



4.2 Recommendations for Future enhancements:

From "Click the Bikes" to AI: The Evolution of Human Verification

We've all had that moment of frustration when we're asked to identify traffic lights or bicycles in a grid of pictures. These challenges, known as **CAPTCHAs**, have been the go-to solution for preventing bots from accessing websites. However, as technology advances, so do the bots, and CAPTCHA is evolving along with it.

The Early Days: A Simple Barrier

When CAPTCHA systems were first introduced, they served a simple purpose: create puzzles that were easy for humans but difficult for machines. These puzzles often involved distorted characters or challenging images. However, as bots became more advanced, they started finding ways to bypass these defenses. What was once a reliable "block" became more easily surmountable.

The Shift: Moving Beyond Static Images to Dynamic Behavior

The major turning point came when CAPTCHA systems began to move away from static image-based puzzles. Instead of just focusing on **what** the user solves, they began to monitor **how** the user solves it. Now, systems track various behaviors, such as:

- **Mouse Movement Patterns:** Is the mouse moving in a natural, unpredictable way, or are the movements straight and robotic?
- **Click Timing:** Does the timing between clicks feel human, with natural pauses, or does it follow an unnatural, consistent pattern?
- **Pacing of Task Completion:** Are there pauses or slight delays when solving the puzzle, or is it completed in a steady, robotic rhythm?

Present Day: A Layered Approach to Security

Today, CAPTCHA is no longer just a simple puzzle; it's part of a broader, **multi-layered security framework** [13]. Think of it as a layered approach to protecting your browsing experience:

1. **First Layer:** Device Fingerprinting – The system identifies the unique attributes of the device accessing the website.



2. **Second Layer:** Behavior Analysis – It tracks how users interact with the site, observing any deviations from normal human behavior.
3. **Third Layer (Advanced CAPTCHA):** The final challenge – This appears only if suspicious activity is detected in the first two layers, providing an additional defense.

The Ever-Evolving Challenge: Outpacing Advanced Bots

Just like airport security systems evolve to address new threats, CAPTCHA systems must continuously adapt to keep up with more sophisticated bots. Modern bots are incredibly adept at imitating human behavior, so CAPTCHA systems need to evolve constantly to stay effective.

The Answer: Self-Updating Intelligence

To effectively combat advanced bots, CAPTCHA can't simply rely on predefined rules—it needs to be **self-learning**. Here's how:

- **Learning from Every Interaction:** The system must improve continuously, analyzing both successful and unsuccessful login attempts.
- **Diverse Data Sources:** CAPTCHA shouldn't just rely on images; it should also take into account behavioral signals like timing, mouse movement patterns, and response structures.
- **Continuous Innovation:** Collaboration between researchers and developers is crucial to ensure that CAPTCHA systems can quickly respond to new, emerging threats.

A Step Toward Invisible Tests

Interestingly, the direction CAPTCHA is heading is toward **invisible verification**. In the future, legitimate users may never even notice CAPTCHA, while bots will be subjected to hidden, multi-layered analyses that challenge them without their knowledge.

Conclusion: From an Annoyance to a Silent Protector

CAPTCHA has transformed over time from a minor nuisance to a sophisticated security tool that works quietly in the background. It's no longer just about identifying images; it's become a complex system that constantly adapts to emerging threats.

The critical question now is whether these systems will continue to stay one step ahead of the ever-evolving bots. The ongoing development of CAPTCHA technology will determine whether it remains an effective, invisible guardian of our digital experiences.



5.Appendix:

We interacted directly with ChatGPT, which responds to specific questions and explains the topic for clear understanding. The answers provided were incorporated into our description to assist with clarification and research. Below is the link of appendix.

<https://chatgpt.com/share/6936597a-ae0c-8012-8116-98634e14c090>



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